

Homework — 2.2.2 Computational Methods

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 2.2.2 lesson. Show traces/working.

Part A — This week's topic (~14 marks)

1. State **one** feature of a problem that makes it possible to solve **computationally**. [1] (AO1)

2. Explain the difference between **decomposition** and **divide and conquer**. [2] (AO1)

3. A weather service displays temperature readings from across the country as a colour-coded **heat map** rather than as a long table of numbers.

(a) State the name of the computational method being used. [1] (AO1)

(b) Give **one** benefit of this method when analysing large data sets. [1] (AO1)

4. A delivery firm uses a **heuristic** to plan the routes for its vans rather than an exact method.

(a) State what is meant by a **heuristic**. [1] (AO1)

(b) Explain **why** a heuristic, rather than an exact algorithm, is appropriate for a route-planning problem such as the Travelling Salesman Problem. **[2]** (AO2)

5. A photo-sharing website processes every uploaded photo in three stages: resize, apply filter, add watermark. Thousands of photos are uploaded every minute.

Explain how **pipelining** could be used to increase the throughput of this system. **[2]** (AO2)

6. A robot explores a maze. It follows a path until it reaches a dead end, then returns to the most recent junction and tries a different path, repeating this until it finds the exit.

(a) State the name of the problem-solving technique the robot is using. **[1]** (AO1)

(b) Explain how this technique works, with reference to the maze scenario. **[2]** (AO2)

7. A supermarket analyses millions of loyalty-card transaction records to discover unexpected patterns and relationships in customers' shopping habits. State the name of this computational method. **[1]** (AO1)

Part B — Retrieval: keep it fresh (~6 marks)

8. (2.2.1) State the defining difference between a **function** and a **procedure**. [1]

9. (2.2.1) A recursive subroutine has no reachable **base case**. State the run-time error that results and explain why it occurs. [2]

10. (2.1.2) State what is meant by **thinking ahead**, and give **two** examples of where it is used when designing a system. [3]
